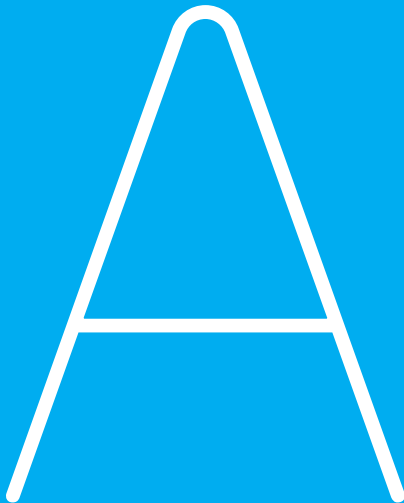


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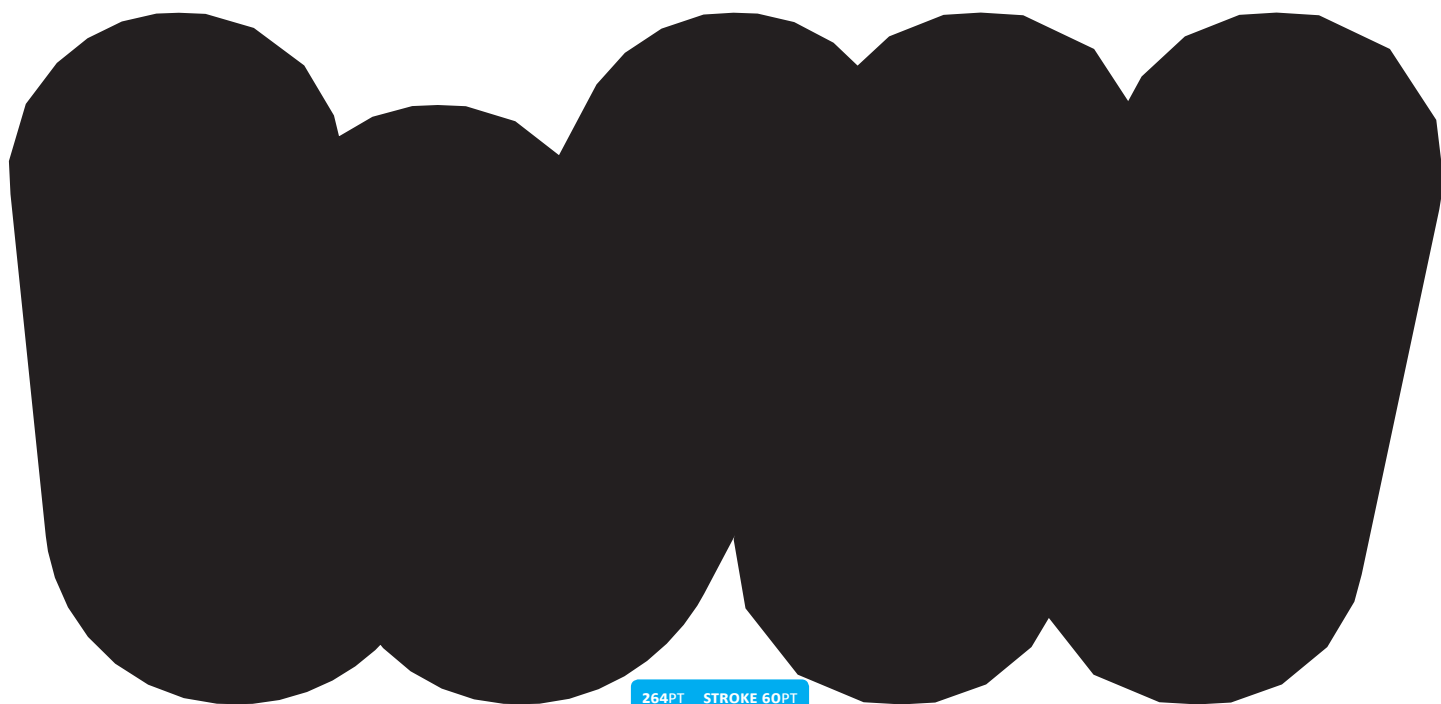
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NINTENDO

87PT STROKE 2PT TRACKING -50



264PT STROKE 60PT

SONY PLAY

82PT STROKE 6PT

STATION

100PT STROKE 14PT SKEW 12°

XBOX 360

90PT STROKE 0PT

MICROSOFT

102PT STROKE 4PT TRACKING -225

BALENCIAGA

67PT STROKE 3PT

DIOR

178PT STROKE 10PT

PRADA

125PT STROKE 0PT

YVES SAINT
LAURENT

75PT STROKE 16PT

92PT STROKE 20PT

VUITTON

100PT STROKE 5PT

HERMES

132PT STROKE 1PT TRACKING -225

DISC
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120PT STROKE OPT TRACKING -75

**DISC
WORLD
NOIR**

120PT STROKE 16PT TRACKING -75

AIR

242PT

WIND

156PT

FRONT

130PT

CUMULUS

88PT

ATMOSPHERE

62PT

CIRROSTRATUS

56PT

UPDOWN

UP / DOWN

PUSH / PULL

LEFT / RIGHT

FORWARD / BACKWARD

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BY JOEL ACHENBACH PHOTOGRAPHS BY PATRICK MCFEELEY

IN THE TEETH OF **JAWS**

IMAGINE SKIING DOWN THE FRONT OF A FIVE-STORY BUILDING. THAT'S **LAIRD HAMILTON**'S IDEA OF FUN, PLUNGING SOME 50 VERTICAL FEET ON A WAVE FACE THAT ONLY A HANDFUL OF SURFERS DARE RIDE. THE STRENGTH OF SUCH WAVES HAS EARNED THIS HAWAIIAN SURFING SPOT ITS NOM DE GUERRE—**JAWS**. THOSE WHO KNOW IT BEST FEAR IT MOST. SAYS HAMILTON: "SOME SAY 'NO FEAR'. I SAY 'BULL'. I ABSOLUTELY FEAR IT. THAT'S WHY I SURF IT."

N

STROKE 1PT

IN **PHYSICAL COSMOLOGY**, THE TERM **BIG BANG** HAS SEVERAL RELATED MEANINGS. IT IS A COSMOLOGICAL MODEL IN WHICH *THE UNIVERSE HAS BEEN EXPANDING FOR AROUND 13.7 BILLION YEARS (13.7 GA), STARTING FROM A TREMENDOUSLY DENSE AND HOT STATE.* THE TERM IS ALSO USED IN A NARROWER SENSE TO DESCRIBE THE FUNDAMENTAL 'FIREBALL' THAT ERUPTED AT OR CLOSE TO TIME $t=0$ IN THE HISTORY OF THE UNIVERSE. OBSERVATIONAL EVIDENCE FOR THE BIG BANG INCLUDES **THE ANALYSIS OF THE SPECTRUM OF LIGHT FROM GALAXIES**, WHICH REVEAL A SHIFT TOWARDS LONGER WAVELENGTHS PROPORTIONAL TO EACH GALAXY'S DISTANCE IN A RELATIONSHIP DESCRIBED BY **HUBBLE'S LAW**. COMBINED WITH THE ASSUMPTION THAT OBSERVERS LOCATED ANYWHERE IN THE UNIVERSE WOULD MAKE SIMILAR OBSERVATIONS (**THE COPERNICAN PRINCIPLE**), THIS SUGGESTS THAT SPACE ITSELF IS EXPANDING. EXTRAPOLATION OF THIS EXPANSION BACK IN TIME YIELDS A STATE IN THE DISTANT PAST IN WHICH THE UNIVERSE WAS IN A STATE OF IMMENSE DENSITY AND TEMPERATURE. THIS HOT, DENSE STATE IS THE **KEY PREMISE** OF THE BIG BANG. THEORETICAL WORK FOR THE BIG BANG COMES FROM MATHEMATICAL MODELS, CALLED **FRIEDMANN MODELS**, WHICH SHOW THAT A BIG BANG IS CONSISTENT WITH GENERAL RELATIVITY AND WITH THE COSMOLOGICAL PRINCIPLE, WHICH STATES THAT THE PROPERTIES OF THE UNIVERSE SHOULD BE **INDEPENDENT OF POSITION OR ORIENTATION**. THE THEORY OF BIG BANG NUCLEOSYNTHESIS PREDICTS THE RATES AT WHICH VARIOUS LIGHT ELEMENTS ARE CREATED IN MODELS OF THE EARLY UNIVERSE AND GIVES RESULTS THAT ARE GENERALLY CONSISTENT WITH OBSERVATIONS. THE BIG BANG MODEL ALSO PREDICTS **THE COSMIC MICROWAVE BACKGROUND RADIATION (CMB)**, A BACKGROUND OF WEAK MICROWAVE RADIATION FILLING THE WHOLE UNIVERSE. THE DISCOVERY OF THE CMB IN 1964 LED TO GENERAL ACCEPTANCE AMONG PHYSICISTS THAT THE BIG BANG IS THE BEST MODEL FOR THE ORIGIN AND EVOLUTION OF THE UNIVERSE. THE BIG BANG THEORY DEVELOPED FROM OBSERVATIONS OF THE STRUCTURE OF THE UNIVERSE AND FROM THEORETICAL CONSIDERATIONS. IN 1912 **VESTO SLIPHER** MEASURED THE FIRST **DOPPLER SHIFT** OF A "SPIRAL NEBULA" (*SPIRAL NEBULA IS THE OBSOLETE TERM FOR SPIRAL GALAXIES*), AND SOON DISCOVERED THAT ALMOST ALL SUCH NEBULAE WERE RECEDING FROM EARTH. HE DID NOT GRASP THE COSMOLOGICAL IMPLICATIONS OF THIS FACT, AND INDEED AT THE TIME IT WAS HIGHLY CONTROVERSIAL WHETHER OR NOT THESE NEBULAE WERE "ISLAND UNIVERSES" OUTSIDE OUR **MILKY WAY**. TEN YEARS LATER, **ALEXANDER FRIEDMANN**, A RUSSIAN COSMOLOGIST AND MATHEMATICIAN, DERIVED *THE FRIEDMANN EQUATIONS FROM ALBERT EINSTEIN'S EQUATIONS OF GENERAL RELATIVITY*, SHOWING THAT THE UNIVERSE MIGHT BE EXPANDING IN CONTRAST TO THE STATIC UNIVERSE MODEL ADVOCATED BY **EINSTEIN**. IN 1924, **EDWIN HUBBLE'S** MEASUREMENT OF THE GREAT DISTANCE TO THE NEAREST SPIRAL NEBULAE SHOWED THAT THESE SYSTEMS WERE INDEED OTHER GALAXIES. INDEPENDENTLY DERIVING FRIEDMANN'S EQUATIONS IN 1927, **GEORGES LEMAITRE**, A BELGIAN ROMAN CATHOLIC PRIEST, PREDICTED THAT THE RECESION OF THE NEBULAE WAS DUE TO THE EXPANSION OF THE UNIVERSE. IN 1931 **LEMAITRE** WENT FURTHER AND SUGGESTED THAT THE UNIVERSE BEGAN AS A SIMPLE "PRIMEVAL ATOM", ECHOING PREVIOUS SPECULATIONS ABOUT THE COSMIC EGG ORIGIN OF THE UNIVERSE. STARTING IN 1924, HUBBLE PAINSTAKINGLY DEVELOPED A SERIES OF DISTANCE INDICATORS, THE FORERUNNER OF THE COSMIC DISTANCE LADDER, USING THE 100 INCH **HOOVER TELESCOPE** AT

STYLISTIC SET 1

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STYLISTIC SET 2

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